

EBU6408 Sensors and Radio Frequency Identification (RFID)

Lecture 2: RFID Components

Part 1: Readers

Dr Fatma Benkhelifa

PhD, Lecturer (Assistant Professor), SIEEE, Editor of IEEE Communications Letters

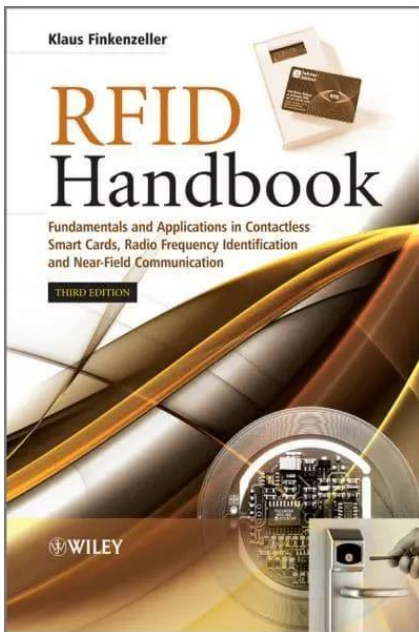
e-mail: f.benkhelifa@qmul.ac.uk

website: [Fatma Benkhelifa - EECS \(qmul.ac.uk\)](http://Fatma Benkhelifa - EECS (qmul.ac.uk))



Textbooks

Klaus Finkenzeller, “RFID Handbook”, Wiley, Third Edition, 2010.



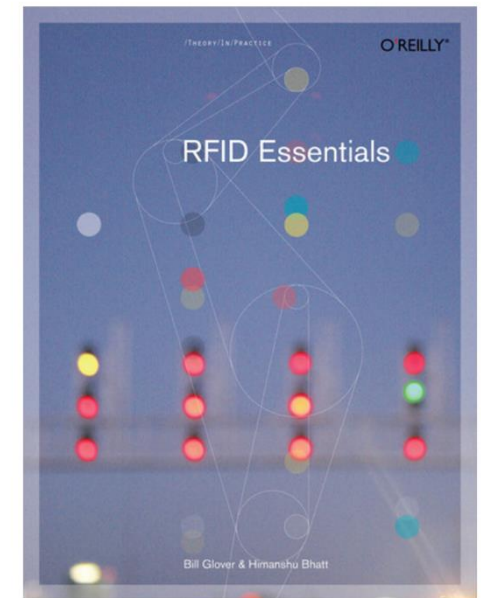
[PDF File](#)

Harvey Lehpamer, “RFID Design Principles”, Artech House, 2012.



[QMUL Library link](#)

Bill Glover and Himanshu Bhatt, “RFID Essentials”, O'Reilly Media Inc, 2006.

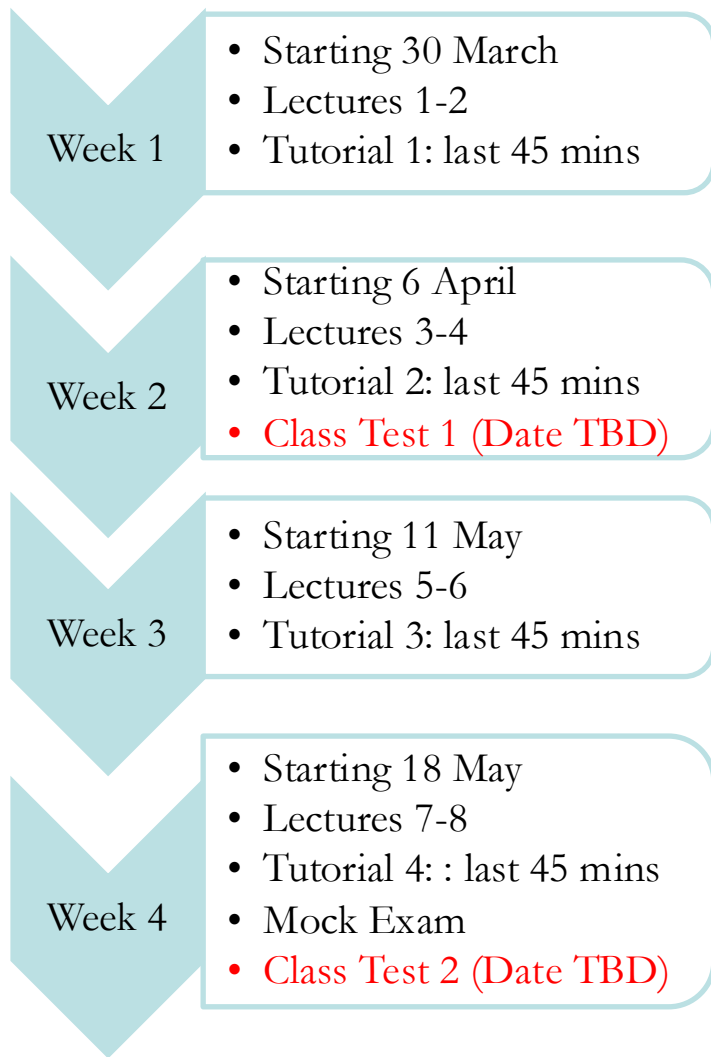


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Evaluation Rules???

- **Two class tests:**
 - **Test 1:** Covering the material delivered in the first half of module **(5%)**
 - **Test 2:** covering the material delivered in the second half of module **(10%)**
 - **Allowed: : Handwriting double-faced A4-sized cheating sheet**
- **Coursework:** Work in groups of 4/5 students.
 - **Lab 1:** Introduction to Arduino **(Bonus Mark)**
 - ✓ Submission of written report on hardware design and programming code.
 - **Lab 2:** RFID projects using Arduino kits **(5%)**
 - ✓ Submission of written report on hardware design and programming code and **5-min video demonstration**
 - **Mini-project:** using practical knowledge from labs and chatbot support **(10%)**
 - ✓ Submission of written report on hardware design and programming code and **10-min video demonstration**
- **Final exam:** covering all teaching weeks **(70%)**
 - **Allowed: Handwriting double-faced A4-sized cheating sheet**
- A couple of QM+ online quizzes on teaching lectures **(Bonus Mark)**

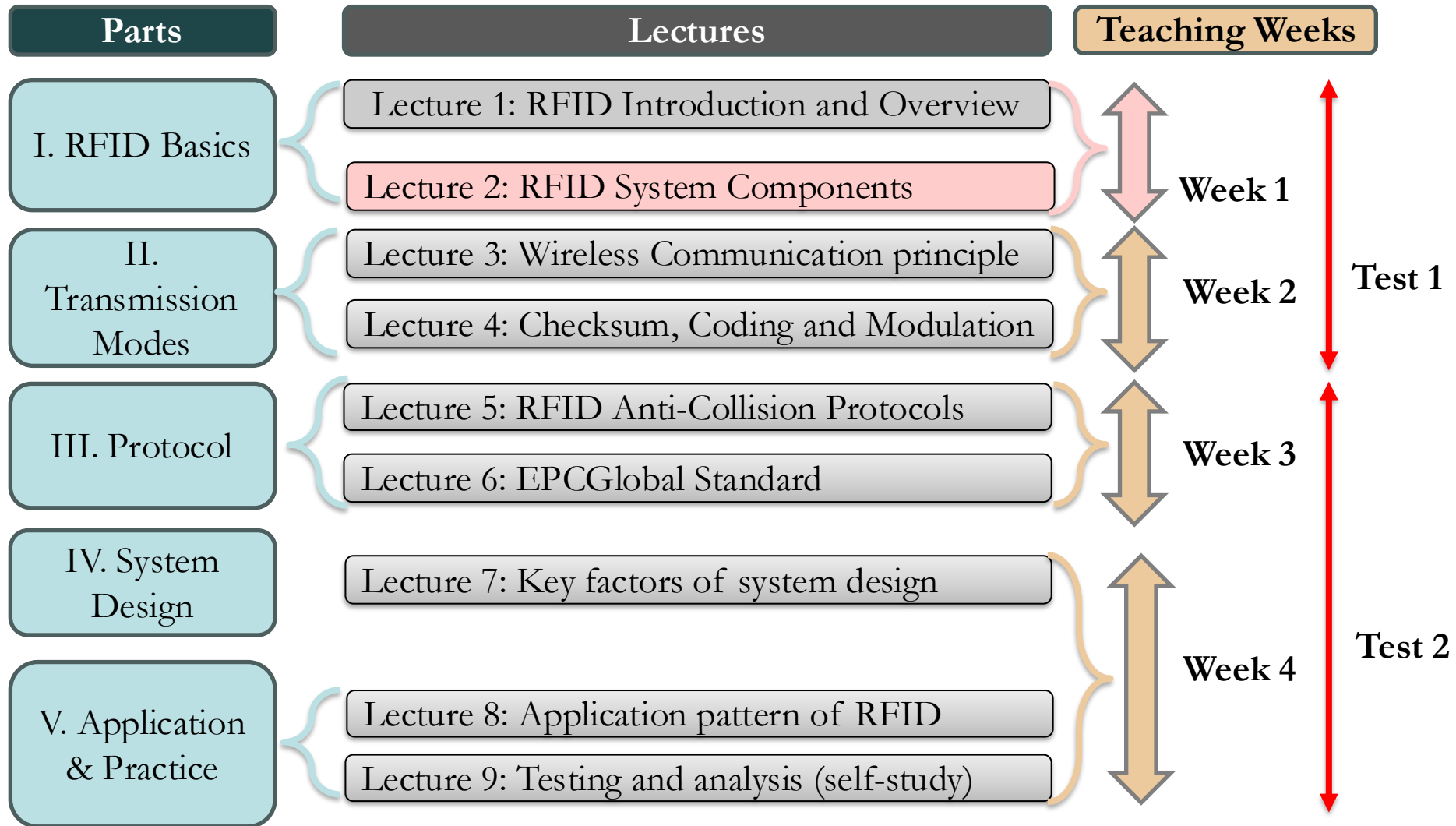
Module Activities



- Labs (Arduino):
 - Two lab sessions will take place in between the teaching weeks.
 - Work in groups (4/5 students) to share the hardware
 - Assignment submission deadlines: usually after lab session.
 - Mini-project due at the end of the term.
- Exact dates will be confirmed later on.

QM+ Quizzes will take place during term time for bonus marks

Module Content



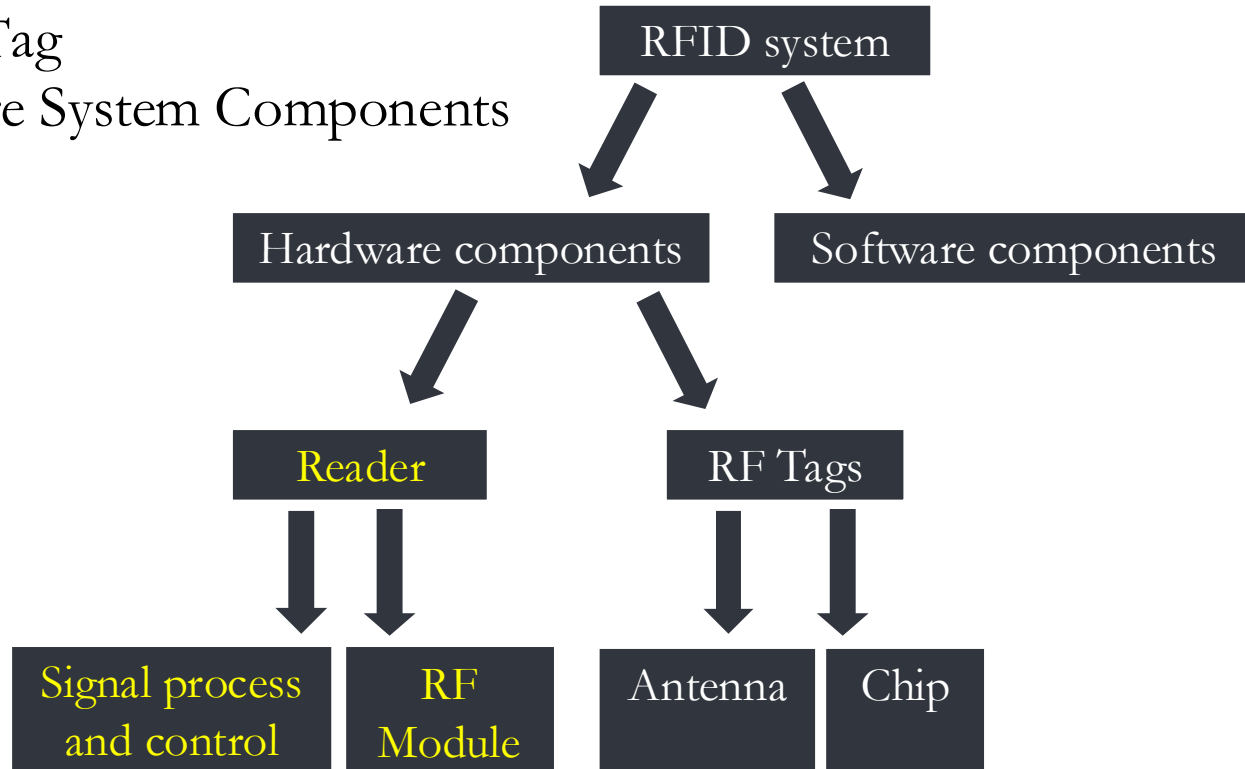
Recap

In last session, we learnt about:

- ❖ What is an RFID
- ❖ Types of automatic identification technology
- ❖ Main features of RFID
- ❖ Core technology of RFID
- ❖ Brief History of RFID
- ❖ Development trends of RFID
- ❖ RFID and Internet of Things

Outline

- ❖ Reader
- ❖ RFID Tag
- ❖ Software System Components



Section

RFID SYSTEM COMPONENTS

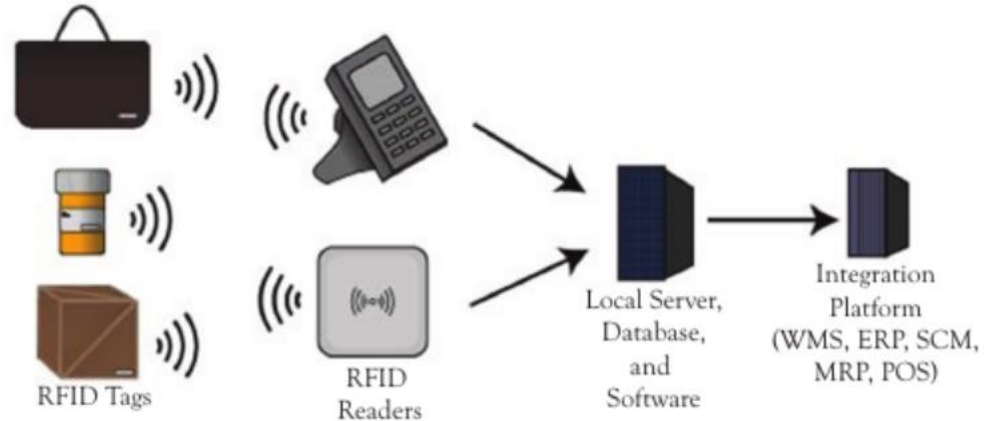
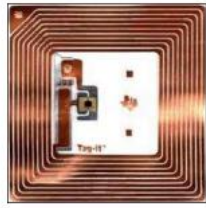


What Makes Up an RFID System?

- RFID systems are comprised of several key components:
 - RFID Reader: The device that communicates with the RFID tag, sends interrogation signals, and receives data.
 - RFID Tags: Devices that store and transmit information to the reader. Tags can be active, passive, or semi-passive.
 - Antenna: Both the reader and the tag contain antennas for signal transmission and reception.
 - Middleware: Software used to process data between the RFID system and enterprise applications.
- Each component plays a unique role in enabling RFID functionality and communication.

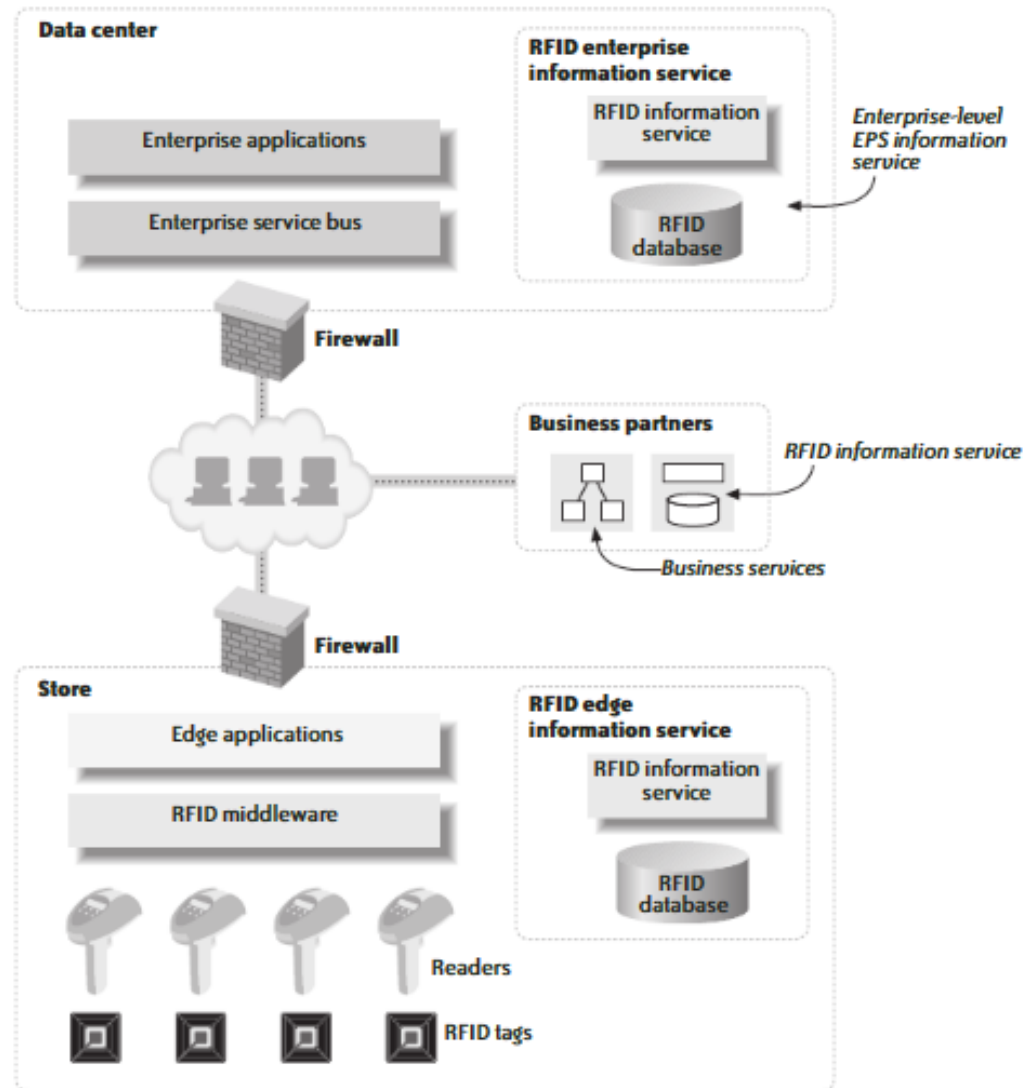


Operating Principle of RFID Systems



- RFID typically consists of
 - A computer
 - A database
 - RFID software
 - A reader, which receives signals from the RFID and identifies and processes the information contained in one or more tags
 - RFID middleware
 - an antenna, used to transmit signals between the reader and the RFID tag.
 - An RFID tag (transponders), which contains the data of the element to identify
- The different for analysed designations of the tags and readers may vary, depending on applications, contexts, and their hardware and software resources.
- The readers are themselves connected to servers for data processing at middleware and application levels is, archiving and traceability

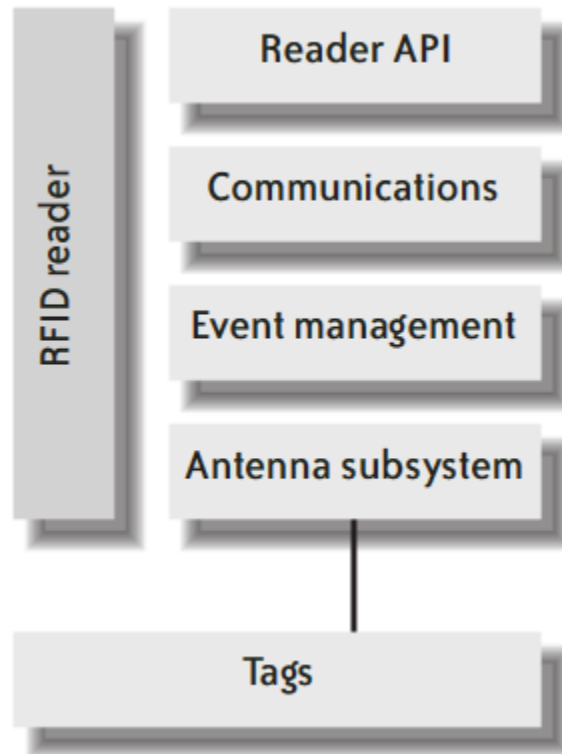
Typical RFID Components in Retail Store



RFID READER



Readers



- ◆ RFID readers, also called interrogators, are used to recognize the presence of nearby RFID tags.
- ◆ An RFID reader transmits RF energy through one or more antennas.
- ◆ An antenna in a nearby tag picks up this energy, and the tag then converts it into electrical energy via induction.
- ◆ This electrical energy is sufficient to power the semiconductor chip attached to the tag antenna, which stores the tag's identity.
- ◆ The tag then sends the identity back to the reader by raising and lowering the resistance of the antenna in a kind of Morse code.

RFID Reader



Reader, a significant part of hardware components in RFID system, will be introduced via the following six parts:

Reader's Function

Reader's
Components

Reader's
Classification

Signal Processing
and Control Module

Reader's Operation
Specifications

Radio Frequency
Module

Reader's Function



- As the bridge connecting the application layer (middleware) and RFID tags, the reader plays a significant role.
- Generally, in RFID system the functions provided by a reader are usually summarized as follows:

Energy Supply

A reader **provides energy** for RFID tags' work.

Security Assurance

Security assurance in communication, such as encryption and decryption

Communication

Communication between reader and tag;
Communication between reader and application layer (middleware)

Ad-hoc networking ability

Multiple antenna **management**

Interface of middle components

Connecting peripherals

...

Reader Classifications: By Operating Frequency



RFID system's working principle is related to the frequency of RF signals it used.

The higher the work frequency, the further the identification range, the higher the data transferring speed, the faster the signal attenuation, the more sensitive to obstacles.



900 MHz

LF and HF readers, work at distance $< 1\text{m}$, typical work frequencies: 125kHz, 135kHz, 6.78MHz, 13.56MHz and 27.125MHz

UHF (Ultra High Frequency) and SHF (Super High Frequency) readers, work at distance $> 1\text{m}$, typical work frequencies: 433MHz, 860MHz~960MHz, 2.45GHz and 5.8GHz

RF Frequency Table

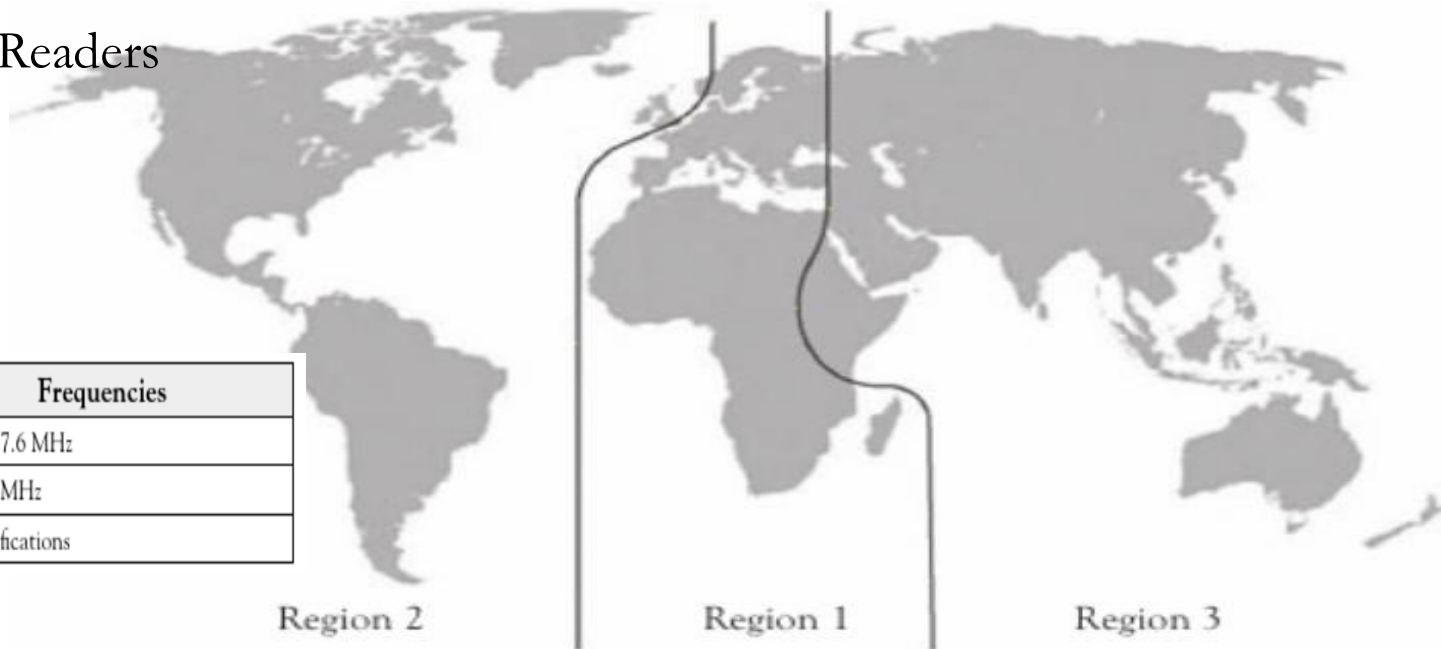
<i>Name</i>	<i>Frequency Range</i>	<i>Applications</i>
Low Frequency (LF)	30 KHz-300 KHz	Time standards, Navigation
Medium Frequency (MF)	300 KHz-3 MHz	AM radio
High Frequency (HF)	3 MHz-30MHz	Amateur radio
Very High Frequency (VHF)	30 MHz-300 MHz	FM/TV broadcasting
Ultra High Frequency (UHF)	300 MHz-3 GHz	Cell phones, WLAN, GPR
Super High Frequency (SHF)	3 GHz-30 GHz	Satellite, 5G mobile
Extremely High Frequency (EHF) “MMW”	30 GHz-300 GHz	Imaging / detection applications, 5G mobile
Tremendously High Frequency (THF)	300 GHz-3 THz	High resolution microwave imaging

Range	LF	HF	UHF	Microwave
Frequency Available	30-300 kHz	3-30 MHz	300-1000 MHz	1-6 Ghz
Used for RFID	125-134 kHz	13.56 MHz	433 MHz & 860-960 MHz	2.4 & 5.8 GHz



Reader Classifications: By Operating Frequency (Cont'D)

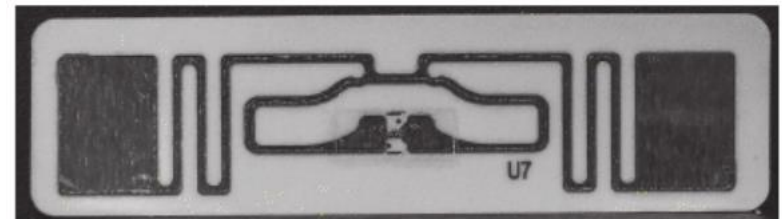
- ◆ UHF passive Readers



Regions	Frequencies
1(Europe and Africa)	865.6–867.6 MHz
2(North and South America)	902–928 MHz
3(Asia Pacific)	No Specifications



Parallax world tag



Avery Dennison AD-236u7
(performs across the 860 to 960
MHz frequency)

Reader Classifications: By Appearance

In practice, taking reader's cost and portability into consideration, the appearance of a reader varies a lot, thus divided into the following 3 types:

Fixed



Wired charging
Highly integrated
Fast set up

Portable



Small volume
Battery charging
Easy to move

Industrial



Industrial related
Capable of integrating
other sensors

Reader's Operational Specifications



To make reader's performance meets the requirements of the application, certain operational specifications should be followed while using the reader.

- **R&W range (read out and write in range)** is one of the key performance metrics that affect the scope of reader applications.
- **Influencing factors:**
 - The way that antenna is coupled
 - The output power of the reader's RF signal
 - The frequency of RF carrier signal



Reader's Operational Specifications (Cont'D)



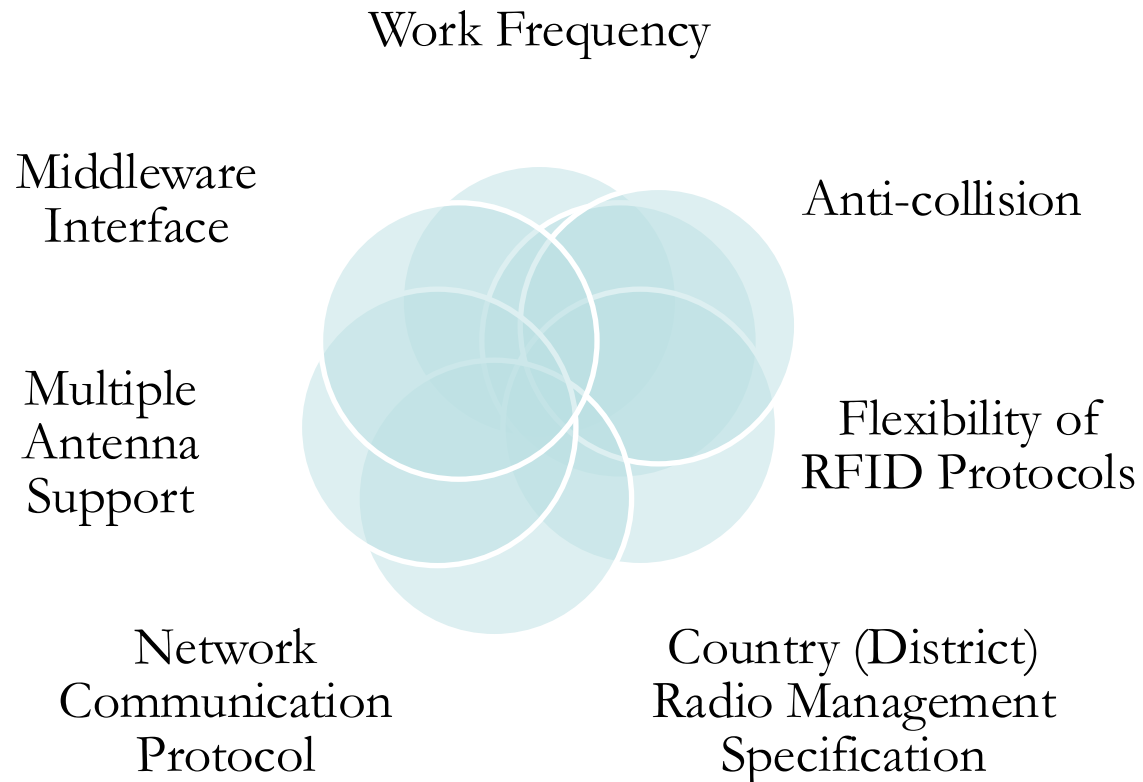
- **Antenna direction**
- When the **polarization direction** of the reader antenna and tag antenna **match**, the recognition range becomes maximum. In practice, mismatch is unavoidable, where measures as followed can be applied:
 - Change the direction of the reader antenna
 - Use multiple redundant antennas
 - Increase the power of the reader antenna
 - Increase sampling rate of the reader antenna
 - Use a ring polarized reader antenna
- **Operation environment condition** (metal interference, electronic noise, multi-tag, multi-reader, etc.)
- **Movement speed of tags**



Reader's Operational Specifications (Cont'D)



Among all operational specifications, here are several aspects that should be paying much more attention:



Reader's Operational Specifications (Cont'D)

- Work frequency
 - Work frequency ↑, data transfer speed ↑, work range ↑.
 - LF system using 125kHz: no more than 400b/s
 - Systems with data transfer speed over 100Kb/s need UHF/microwave
- Anti-collision performance
 - The system's effective data transfer rate in scenario with multiple RF tags reading and writing
- Flexibility of RFID protocols
 - Communication protocol: universal protocol (ISO18000-6C, EPCglobal Class1 Generation2, etc.) or exclusive protocol
- Country (district) radio management specification
 - USA UHF RFID bandwidth range: 902MHz-930MHz
 - China UHF RFID bandwidth range: 840MHz-845MHz, 920MHz-925MHz
 - Maximum transmit power requirements

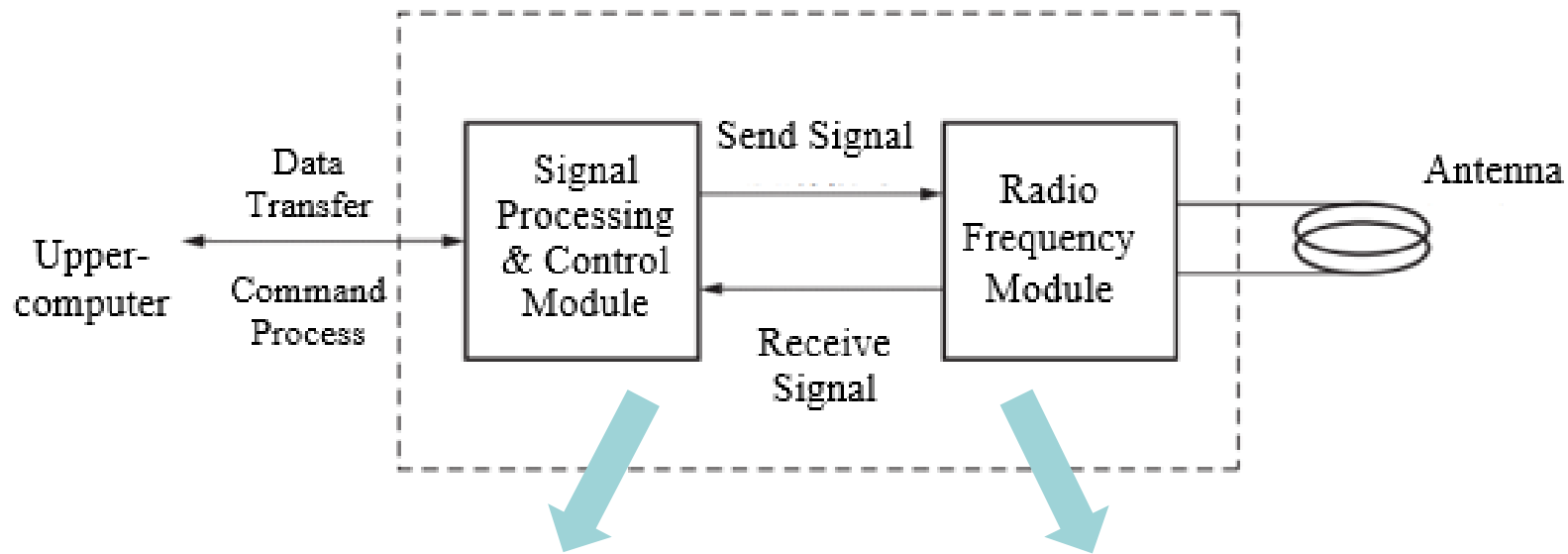


Reader's Operational Specifications (Cont'D)

- Communication protocols
 - Communication interface between reader and computer network system (TCP/IP, 802.11, Ethernet, RS485, etc.)
- Multiple antenna support ability
 - Whether to support multiple antennas for one reader
- Middleware interface
 - Stable performance, easy to use
- Connect external sensor nodes and control circuits
 - Combine RFID and sensor networks
 - Use PLC (programmable logic controller) to perform entry management for warehouses

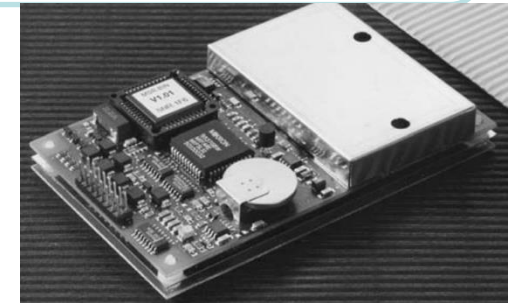


RFID Reader Block Diagram



Known as baseband control module, includes: 1) Microprocessor to perform computing tasks, 2) Digital signal processing chip to encode and decode digital signal

Known as high frequency interface module, includes separate channels: 1) Transmitter signal channel and 2) Receiver signal channel



Signal Processing and Control Module

Communicate with upper computer,
and execute command from it

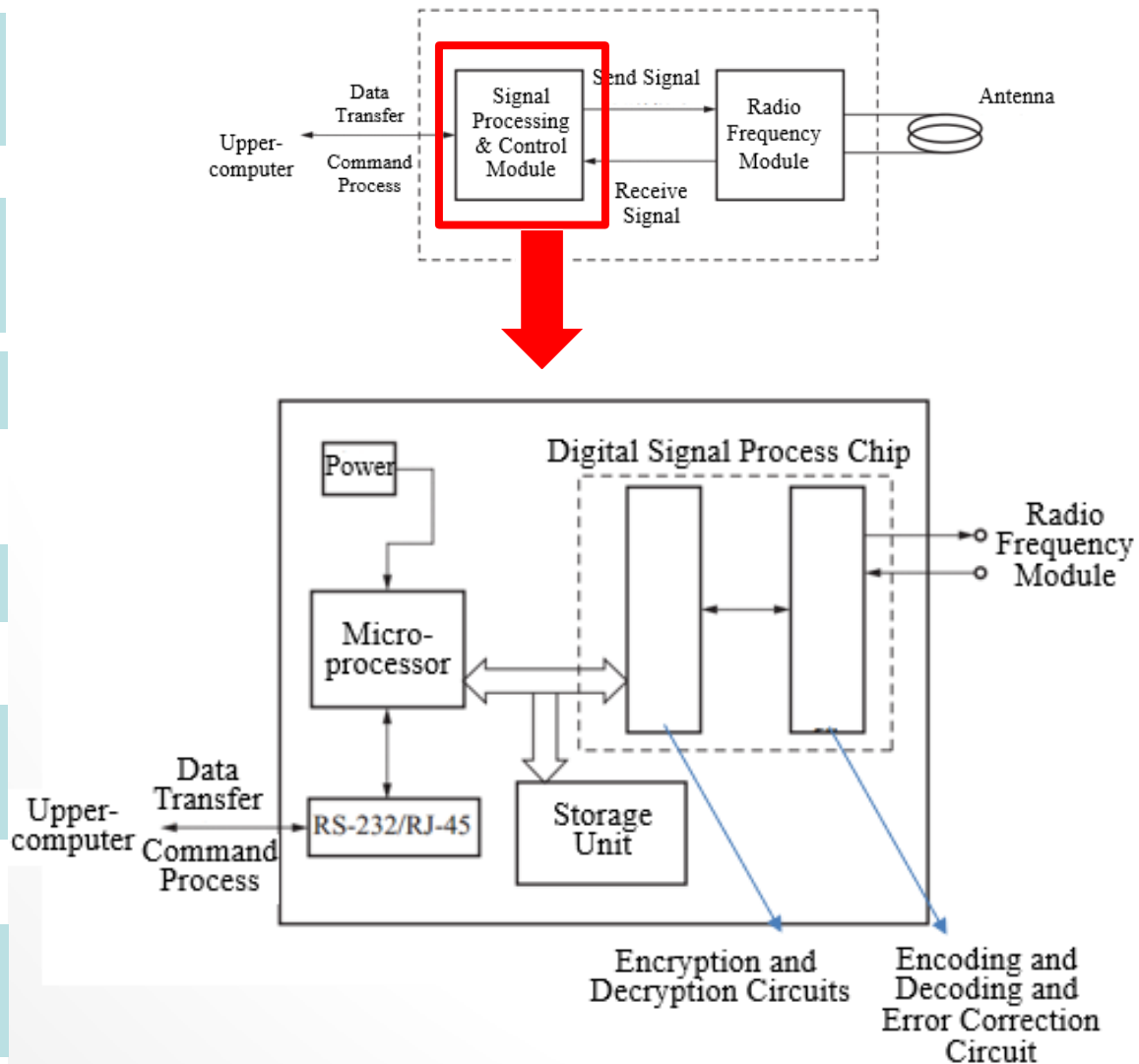
Control communication process
with tags

Encode and decode signal

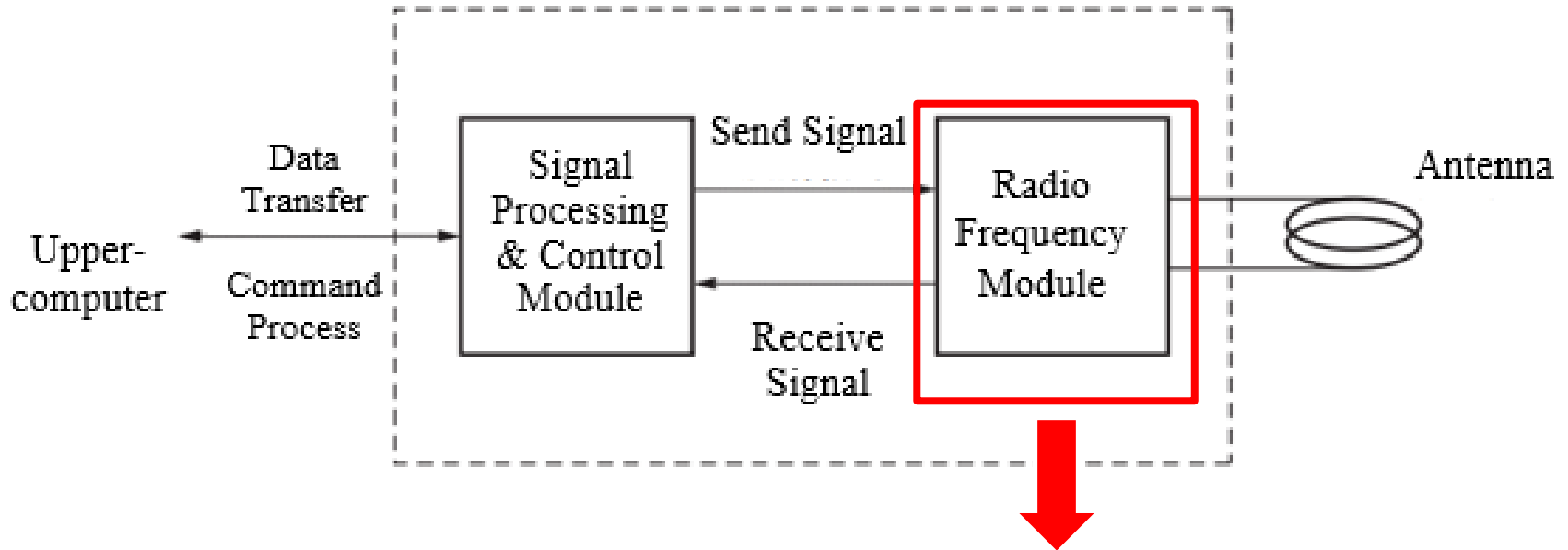
Perform anti-collision algorithm

Encrypt and decrypt the data
transferred between reader and tag

Identity certification between
reader and tag



RF Module



2 Types

- **Inductively Coupled RF Modules:** Ideal for short-range applications, using magnetic fields for communication.
- **Electromagnetic Backscatter Coupled RF Module:** Suitable for longer ranges, leveraging reflected signals for data transfer.

RF Interface

- The reader's RF interface performs the following functions:
 - generation of high-frequency transmission power to activate the transponder and supply it with power;
 - modulation of the transmission signal to send data to the transponder;
 - reception and demodulation of RF signals transmitted by a transponder.
- The RF interface contains two separate signal paths to correspond with the two directions of data flow from and to the transponder.
 - Data transmitted to the transponder travels through the *transmitter arm*.
 - Conversely, data received from the transponder is processed in the *receiver arm*.

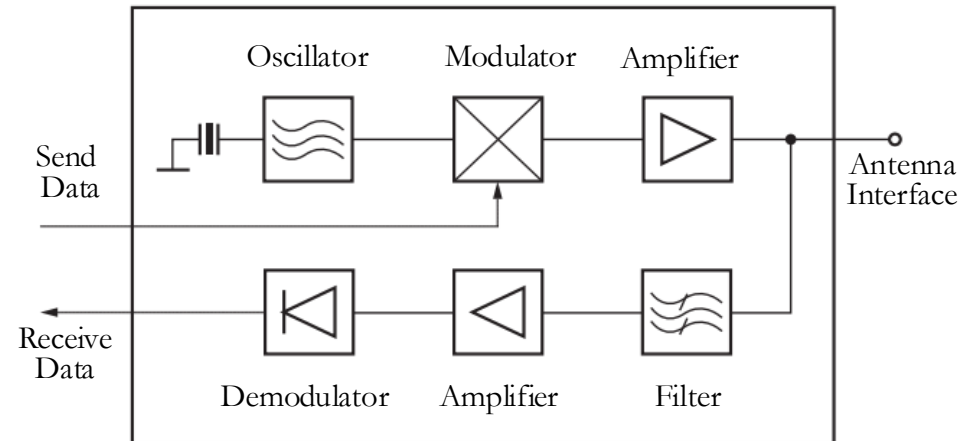
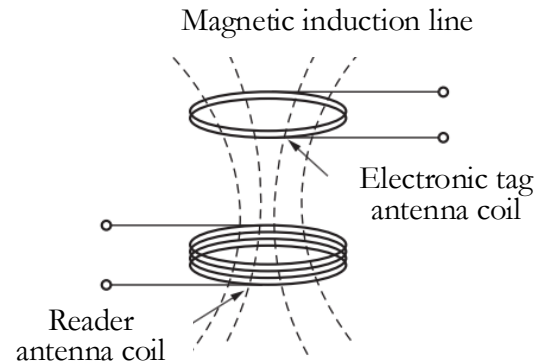


Inductively Coupled RF Modules

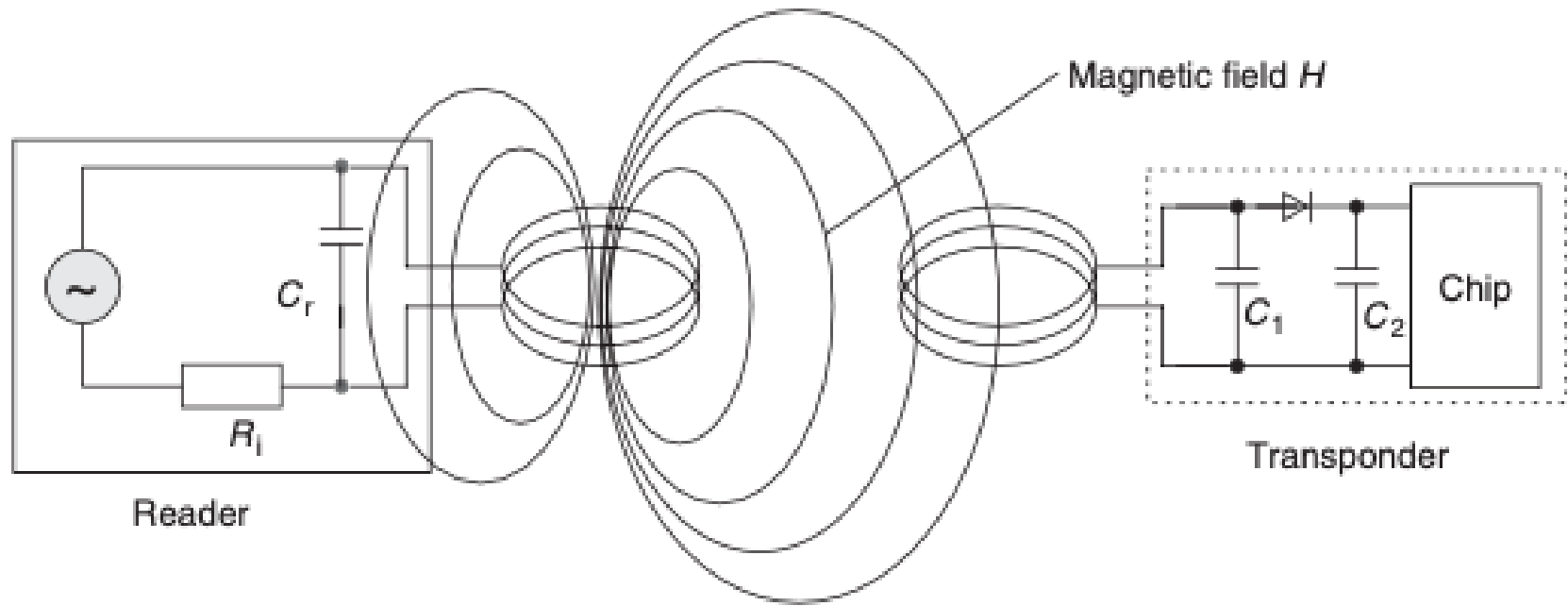
Near-distance LF, HF RFID systems work via inductive couple between reader and RF tag. Tags in this working way are usually **passive**, while gaining energy through inductive couple.

While **sending data to RF tags**, a reader can apply various digital modulation technologies.

While **sending data to a reader**, RF tags usually use load modulation technology, transferring voltage variation in antenna coil of RF tag to the reader antenna.



Inductive Coupling



- ◆ This system typically operates at Low Frequency (LF: 125–134 kHz) or High Frequency (HF: 13.56 MHz).
- ◆ The reader's alternating current generates a magnetic field, which induces a current in the transponder's coil.
- **Limitations:** Works only in close proximity (typically a few centimeters to a meter), depending on the frequency and coil size.

Real Examples



Contactless Payment



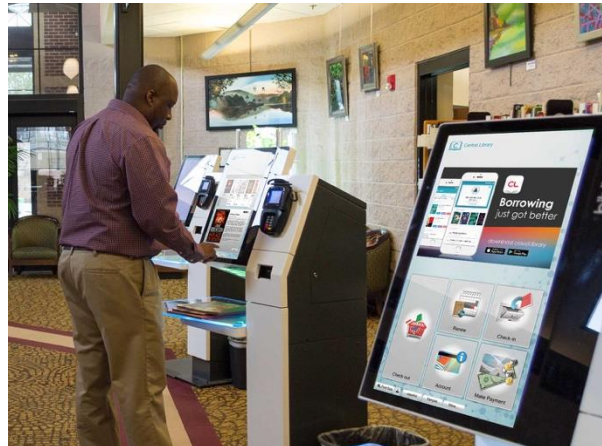
Public Transport and Smart Cards



Hotel Key Cards
& Access Control



Animal Microchip Implants



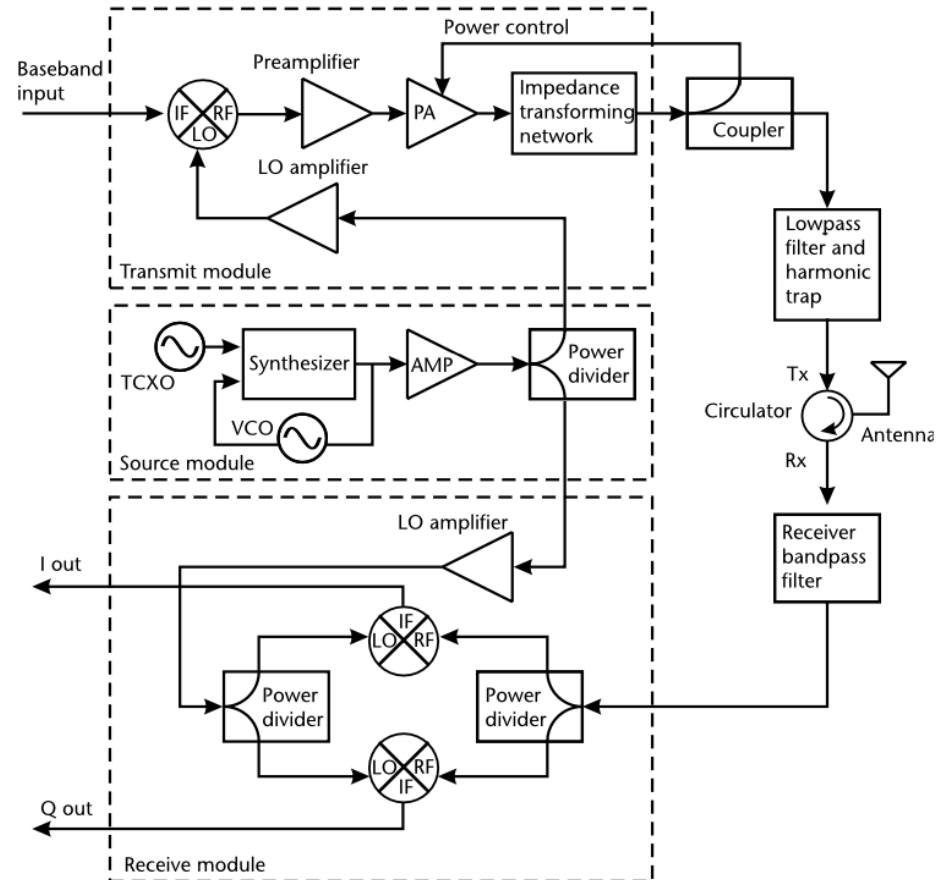
Library and Book Tracking



Car RFID-Based
Ignition Systems

Electromagnetic Backscatter Coupled RF Module

- Long-distance UHF RFID system use electromagnetic backscatter coupled principle between reader and RF tags, **similar to radar**.
- In the system, to **continuously provide energy** for RF tags, the reader must continuously send RF signals.
- The signals sent by reader and returned by tag have the same frequency but **different strength**.
- UHF RF modules can be divided into: **source module, send module and receive module**.

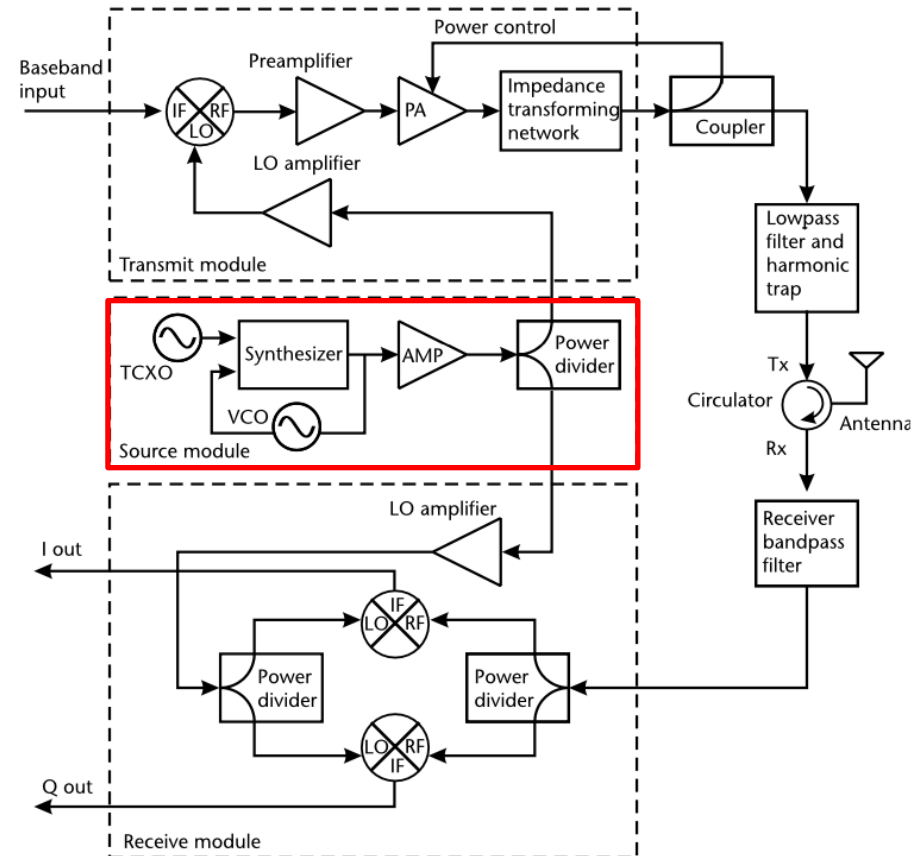


Electromagnetic Backscatter Coupled RF Module

Source Module

Source module provides Local Oscillator for Send Module and Receive Module.

Amplified **carrier signals** are separated into two directions by splitters: to Send/Receive Module.

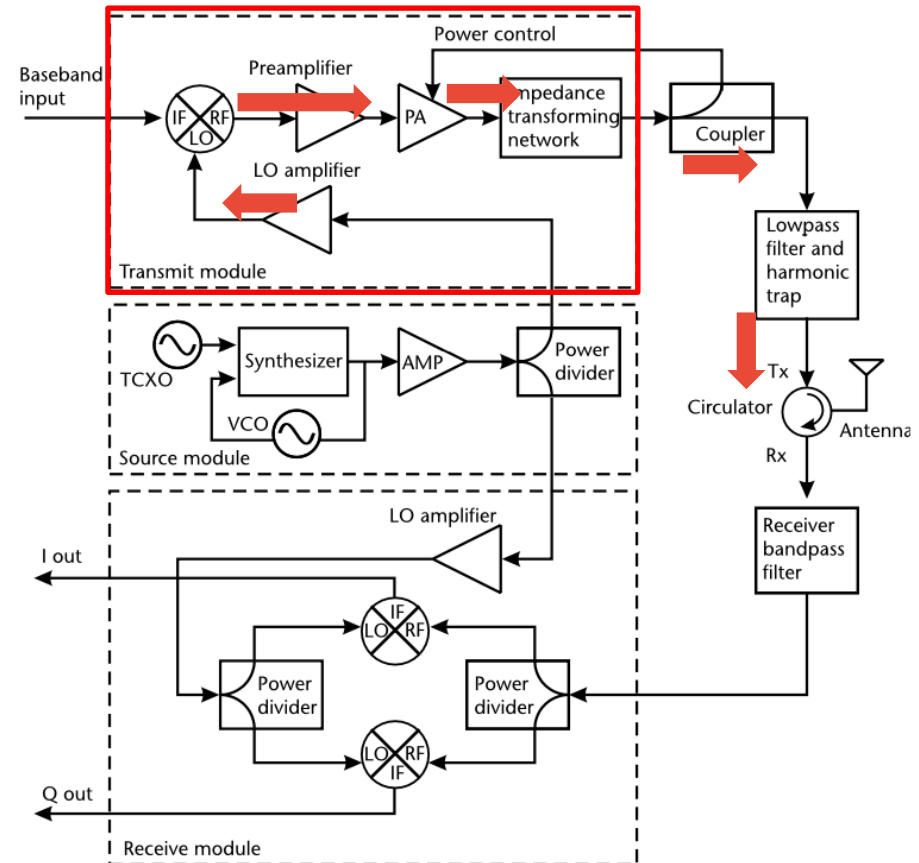
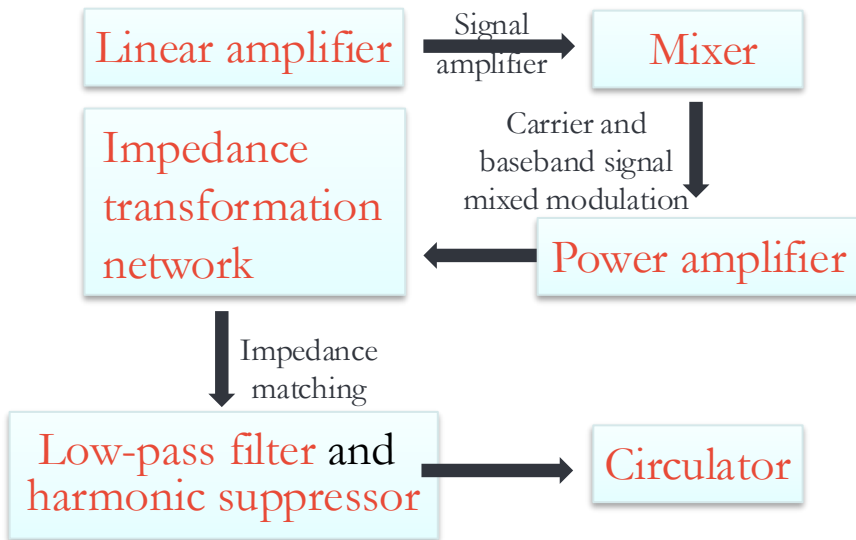


VCO: voltage-controlled oscillator

Electromagnetic Backscatter Coupled RF Module

Send Module

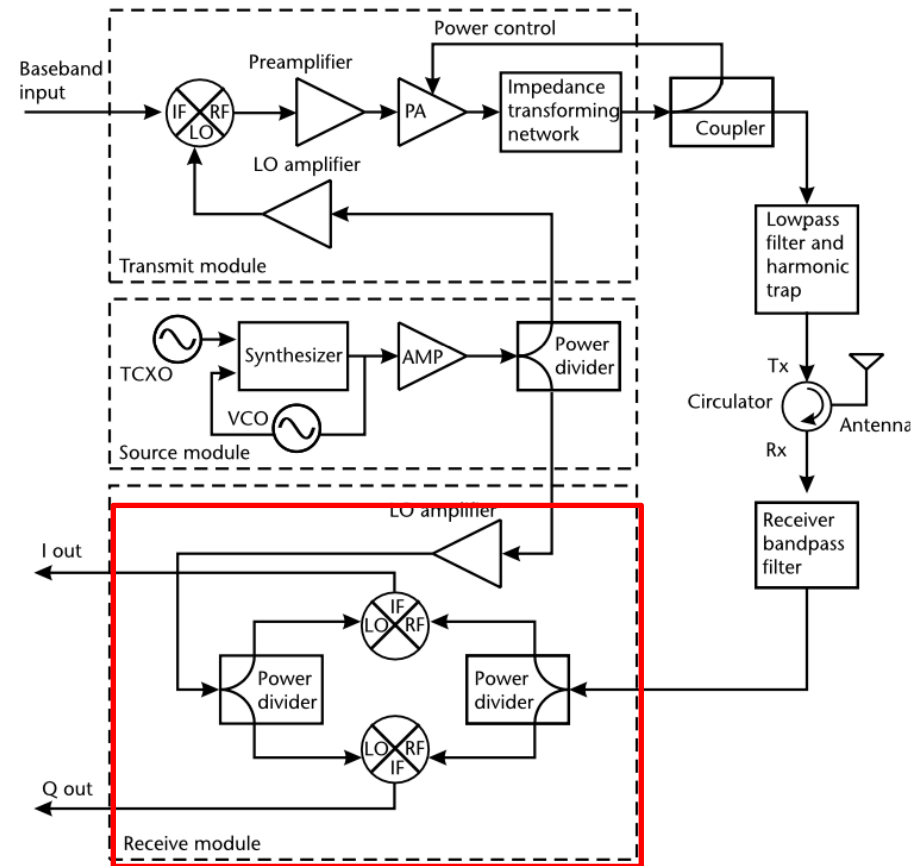
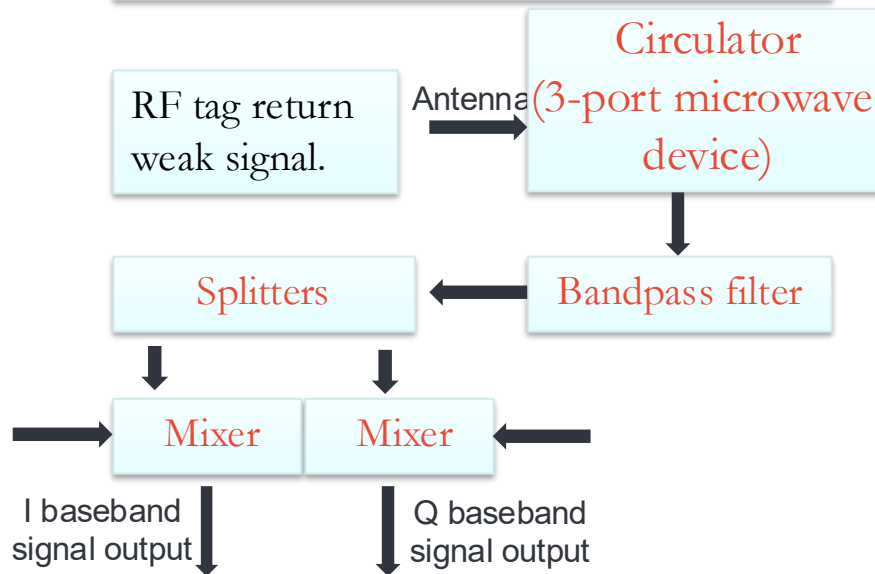
Send Module consists of mixer, local oscillator (LO) amplifier, preamplifier, power amplifier and impedance transformation network.



Electromagnetic Backscatter Coupled RF Module

Receive Module

Receive module consists of a **linear amplifier**, two **splitters** and two **mixers**.



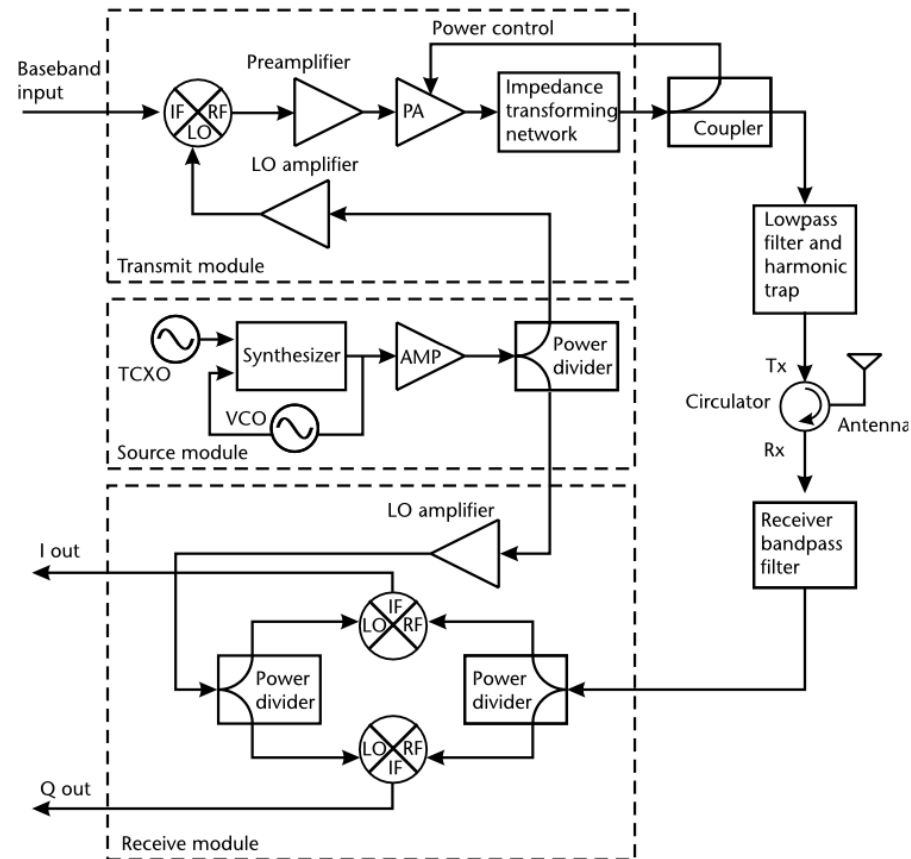
Electromagnetic Backscatter Coupled RF Module

Main Functions of RF Module

Generate high frequency send energy, **activate** RF tags and **provide energy** (passive RF tags)

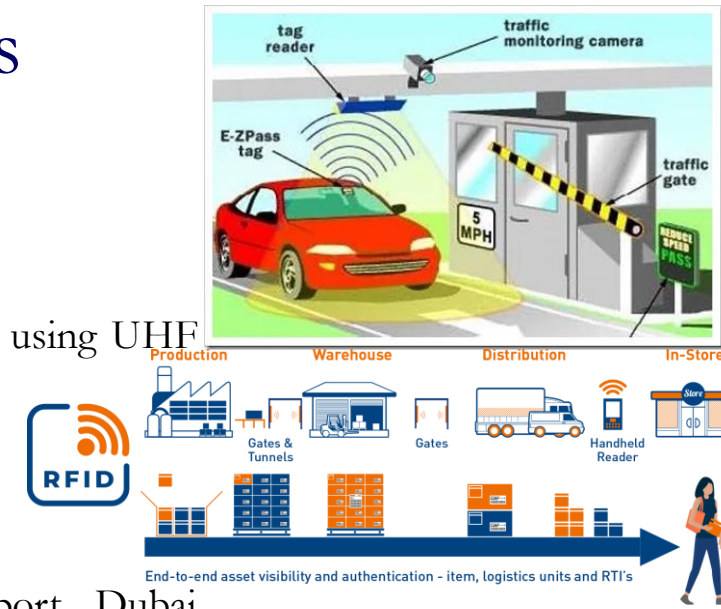
Modulate signal to send, transferring data to RF tags.

Receive and **demodulate** RF signal from RF tags.



Real Examples

- **Electronic Toll Collection (ETC) Systems**
 - **Example:** EZ-Pass (USA), FASTag (India), Telepass (Italy)
- **Supply Chain & Inventory Tracking**
 - **Example:** Walmart, Amazon, Zara, and logistics companies using UHF RFID
- **Retail Self-Checkout Systems**
 - **Example:** Decathlon stores using UHF RFID for checkout
- **Baggage Handling at Airports**
 - **Example:** Delta Airlines, Hong Kong International Airport, Dubai International Airport
- **Race Timing in Marathons & Sports Events**
 - **Example:** RFID timing chips in marathons like the **Boston Marathon**
- **Vehicle Access & Parking Management**
 - **Example:** RFID-based gated parking in offices and malls
- **Livestock & Agriculture Management**
 - **Example:** RFID ear tags used in cattle tracking (e.g., USDA-approved UHF tags)
- **Military & Defense Asset Tracking**
 - **Example:** U.S. Department of Defense's RFID-based logistics (SAVI RFID tags)



Reader Antenna

- The reader antenna establishes a connection between the reader electronics and the electromagnetic wave in the space.
 - HF range: the reader antenna is a coil (like the tag antenna) designed to produce as strong a coupling as possible with the tag antenna.
 - UHF range: reader antennas (like tag antennas) come in a variety of designs. Highly directional, high-gain antennas are used for large read distances.
- One advantage of highly directional antennas is that the reader power often has to be emitted only to the spaces in which the tags that are to be read are located.
- Generally, the higher the gain, the larger the mechanical design of the antenna will be. It follows, therefore, that highly directional antennas are not used for handheld readers.
 - Antennas typically used for handheld readers include patch antennas, half-wave dipoles, and helix antennas.
 - Larger antenna structures can be used for stationary readers; in the UHF range, they usually take the form of arrays.



Reader Antenna (Cont'D)

- All other things being equal, a high-gain antenna will transmit and receive weaker signals farther than a low-gain antenna.
 - **Omnidirectional antennas:** such as dipole antennas, will have lower gain than directional antennas because they distribute their power over a wider area.
 - **Parabolic antennas:** usually have the **highest gain** of any type of antenna but **not really useable in typical RFID applications**, except maybe for microwave RFID readers where a long-range and narrow radiation pattern is required.
 - **A half-wave dipole antenna:** will have a gain of near unity or nearly equal the isotropic antenna.
- Reader antennas may have different requirements depending on whether they are fixed, portable, or mobile readers.



Selection of Antenna for Handhelds

- Portable devices: selection of antennas is **dominated by size and weight constraints**.
 - Read range and polarization are generally less significant than in the case of fixed readers.
 - Efficient use of RF power to maximize battery life is critical.
 - **Highly directive** antennas are useful to reduce power consumption, but are generally physically large.
 - The fact that handheld reader antennas are small and light constrains the antenna gain.
- The gain-size trade-off is important because handheld and portable applications benefit from high antenna gain.
 - Antennas that are less than about $1/4$ of a wavelength in all dimensions cannot achieve more than about 4 dBi of gain.
 - Slightly larger antennas allow up to about 6 dBi of gain, but make the reader large.
- A narrow antenna beam will improve the ability of the user to locate the tag being read by changing the reader orientation and noting the results.
- The narrow beam of a high-gain antenna is often beneficial for a handheld reader, since the user can readily move the beam to cover the area of interest.

Thank You



Dr Fatma Benkhelifa